

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DevOps Lab Experiment – 1

Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup

Overview of Build Automation Tools

Build automation tools help developers streamline the process of building, testing, and deploying software projects. They handle repetitive tasks such as compiling code, managing dependencies, and packaging applications, making development more efficient and less error prone.

Two popular tools in the Java ecosystem are **Maven** and **Gradle**. Both are widely used for managing project builds and dependencies, but they differ in their approach and features.

Maven

Maven is a build automation tool primarily used for Java projects. It uses an XML configuration file called **pom.xml** (Project Object Model) to define *project settings, dependencies, and build steps*

Maven Features

- Predefined project structure and lifecycle phases.
- Automatic dependency management through Maven Central.
- Wide range of plugins for things like testing and deployment.
- Supports complex projects with multiple modules.

Gradle

Gradle is a more modern and versatile build tool that supports multiple programming languages, including Java, Groovy, and Kotlin. It uses a domain-specific language (DSL) for build scripts, written in Groovy or Kotlin.

Gradle Features

- Faster builds thanks to task caching and incremental builds.
- Flexible and customizable build scripts.
- Works with Maven repositories for dependency management.
- Excellent support for multi-module and cross-language projects.
- Integrates easily with CI/CD pipelines

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Difference between Maven and Gradle

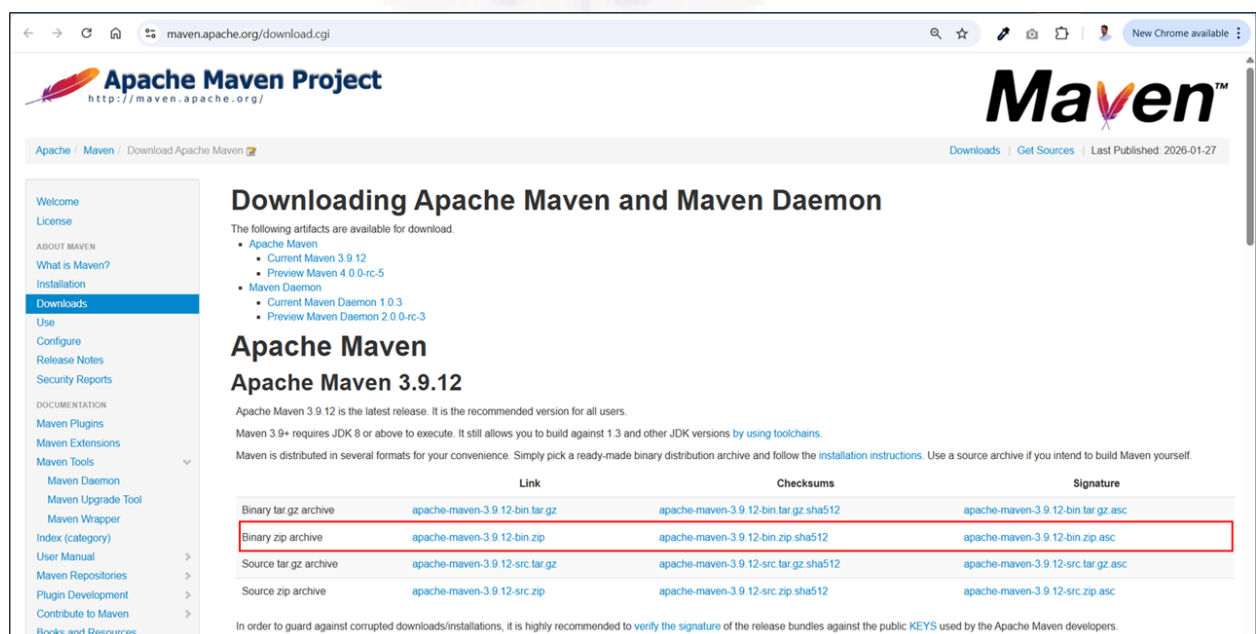
Aspect	Maven	Gradle
Configuration	XML (pom.xml)	Groovy or Kotlin DSL
Performance	Slower	Faster due to caching
Flexibility	Less flexible	Highly customizable
Learning Curve	Easier to pick up	Slightly steeper
Script Size	Verbose	More concise
Dependency Management	Uses Maven Central	Compatible with Maven too
Plugin Support	Large ecosystem	Extensible and versatile

Installation and Environment Setup

Install Maven

1. Download Maven

Go to the [Maven Download Page](https://maven.apache.org/download.cgi) and download the latest binary ZIP file



The screenshot shows the Apache Maven Project download page. The main heading is "Downloading Apache Maven and Maven Daemon". Below this, it lists the following artifacts available for download:

- Apache Maven
 - Current Maven 3.9.12
 - Preview Maven 4.0.0-rc-5
- Maven Daemon
 - Current Maven Daemon 1.0.3
 - Preview Maven Daemon 2.0.0-rc-3

The section "Apache Maven" is expanded, showing "Apache Maven 3.9.12". It states that Apache Maven 3.9.12 is the latest release and is recommended for all users. It also mentions that Maven 3.9+ requires JDK 8 or above to execute.

The table below lists the download links, checksums, and signatures for the available artifacts:

	Link	Checksums	Signature
Binary tar.gz archive	apache-maven-3.9.12-bin.tar.gz	apache-maven-3.9.12-bin.tar.gz.sha512	apache-maven-3.9.12-bin.tar.gz.asc
Binary zip archive	apache-maven-3.9.12-bin.zip	apache-maven-3.9.12-bin.zip.sha512	apache-maven-3.9.12-bin.zip.asc
Source tar.gz archive	apache-maven-3.9.12-src.tar.gz	apache-maven-3.9.12-src.tar.gz.sha512	apache-maven-3.9.12-src.tar.gz.asc
Source zip archive	apache-maven-3.9.12-src.zip	apache-maven-3.9.12-src.zip.sha512	apache-maven-3.9.12-src.zip.asc

In order to guard against corrupted downloads/installations, it is highly recommended to verify the signature of the release bundles against the public KEYS used by the Apache Maven developers.

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2. Extract the ZIP file

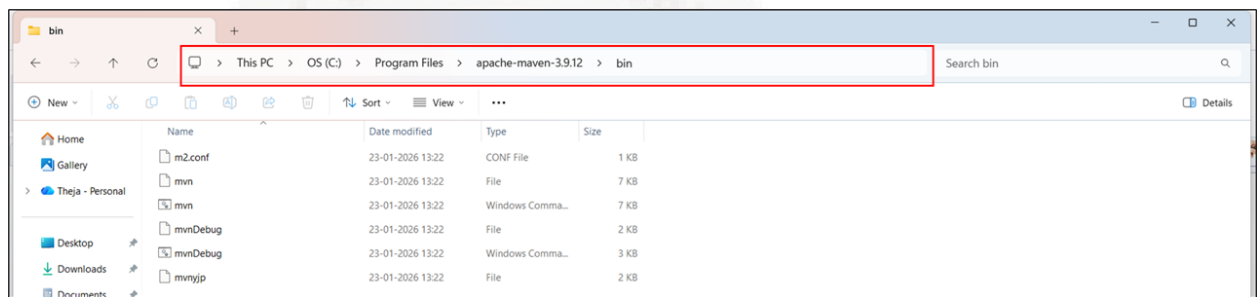
Right-click the downloaded ZIP file and select **Extract All**, or extract it using any compatible tool such as WinRAR or 7-Zip.

3. Move the folder

After extraction, move the extracted Maven folder (usually named **apache-maven-x.x.x**) to a convenient directory like C:\Program Files\.

4. Navigate to the bin Folder

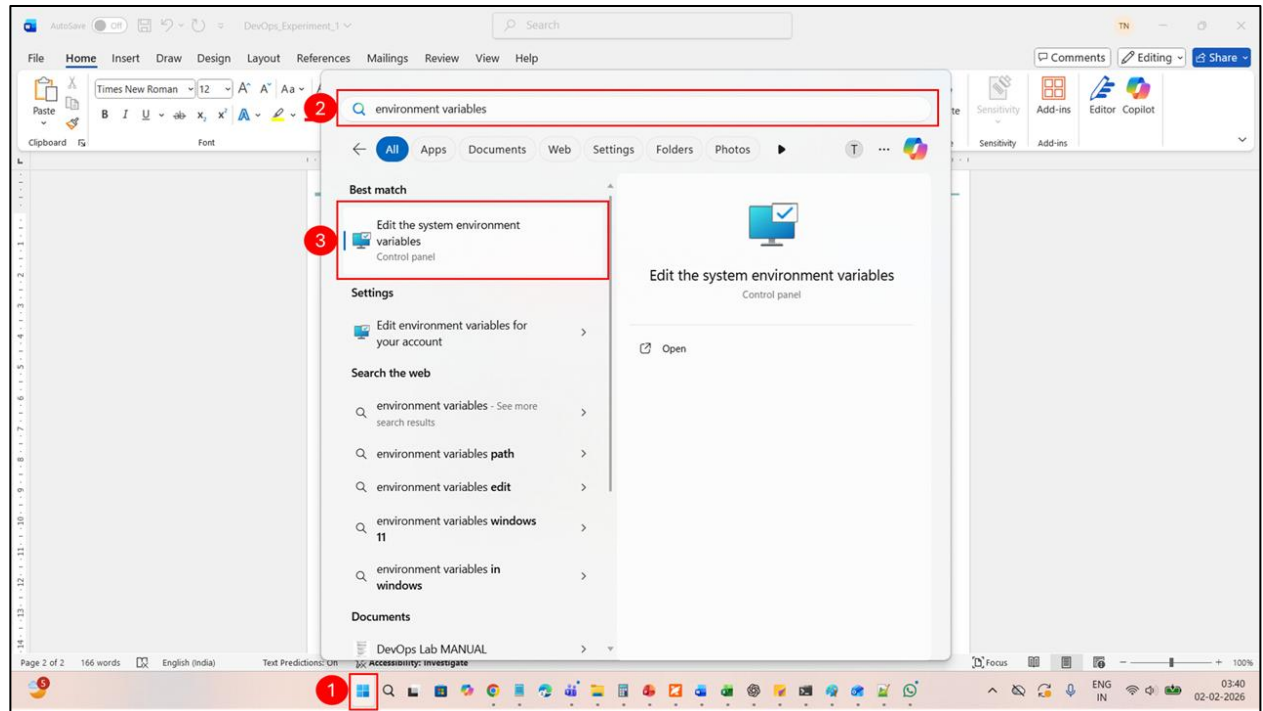
- Open the Maven folder
- Navigate to the bin folder inside it
- Copy the path from the File Explorer address bar (e.g., C:\Program Files\apache-maven-x.x.x\bin)



5. Set Environment Variables

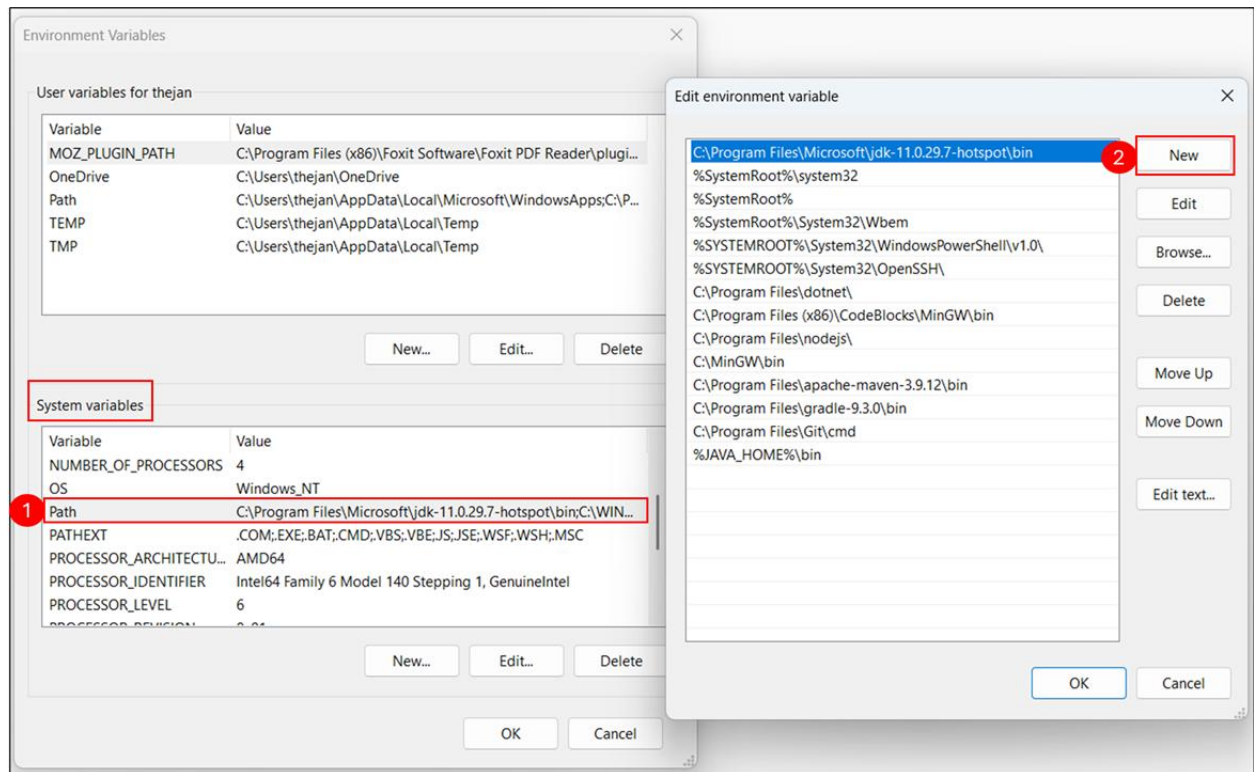
- Open the Start Menu, search for **Environment Variables**, and select **Edit the system environment variables**.

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- Click Environment Variables.
- Under System Variables:
 - Locate **Path**, **double-click it**, and click New.
 - Paste the full path to the bin folder of your Maven directory (e.g., C:\Program Files\apache-maven-x.x.x\bin).

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6. Save the Changes

Click **OK** to close the windows and save your changes.

7. Verify the Installation

Open Command Prompt and run: **mvn -v**. If Maven is correctly installed, it will display the version number.

```
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

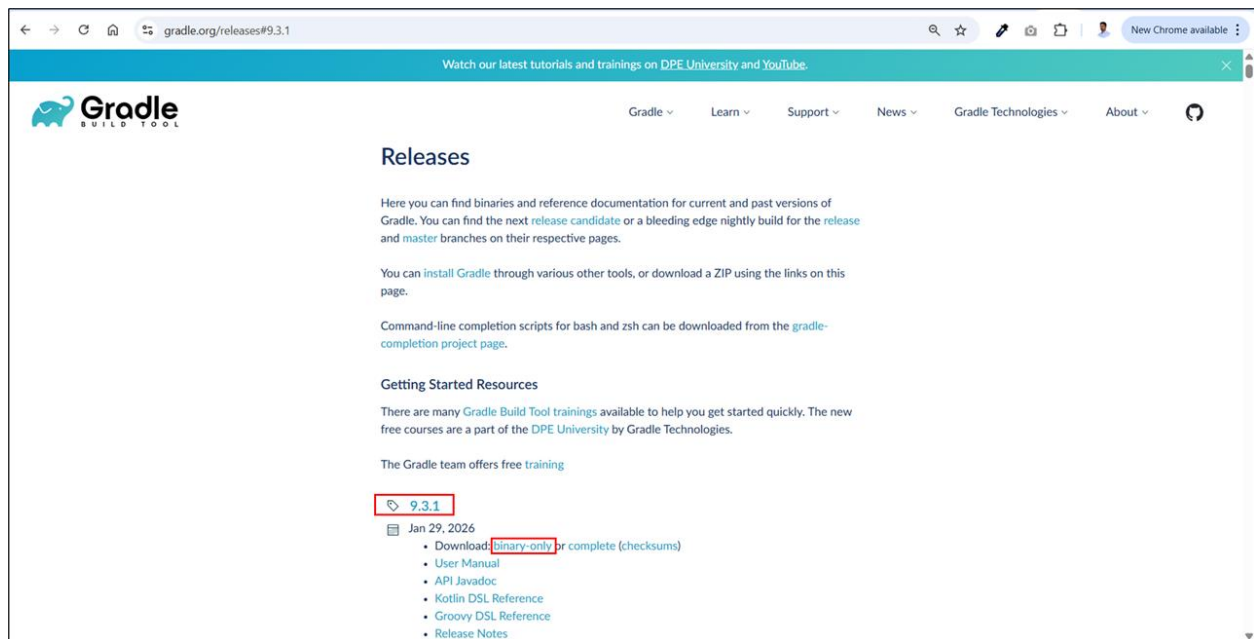
C:\Users\thejan>mvn -v
Apache Maven 3.9.12 (848fbb4bf2d427b72bdb2471c22fced7ebd9a7a1)
Maven home: C:\Program Files\apache-maven-3.9.12
Java version: 11.0.29, vendor: Microsoft, runtime: C:\Program Files\Microsoft\jdk-11.0.29.7-hotspot
Default locale: en_IN, platform encoding: Cp1252
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```


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Install Gradle

1. Download Gradle:

Visit the [Gradle Downloads Page](https://gradle.org/releases) and download the latest binary ZIP file.



2. Extract the ZIP File:

Right-click the downloaded ZIP file and select **Extract All** or extract it using any compatible tool such as WinRAR or 7-Zip.

3. Move the Folder:

After extraction, move the extracted Gradle folder (usually named gradle-x.x.x) to a convenient directory like [C:\Program Files\](#).

4. Navigate to the bin Folder:

- Open the Gradle folder, then navigate to the bin folder inside.
- Copy the path from the File Explorer address bar (e.g., C:\Program Files\gradlex.x\bin).

5. Set Environment Variables:

- Open the Start Menu, search for **Environment Variables**, and select **Edit the system environment variables**.
- Click Environment Variables.
- Under System Variables:

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- Locate **Path**, **double click** it and click **New**.
- Paste the full path to the bin folder of your Gradle directory (e.g., C:\Program Files\gradle-x.x.x\bin).

6. Save the Changes

Click OK to close the windows and save your changes.

7. Verify the Installation

Open Command Prompt and run: **gradle -v** If it shows the Gradle version, the setup is complete

```
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thejan>gradle -v

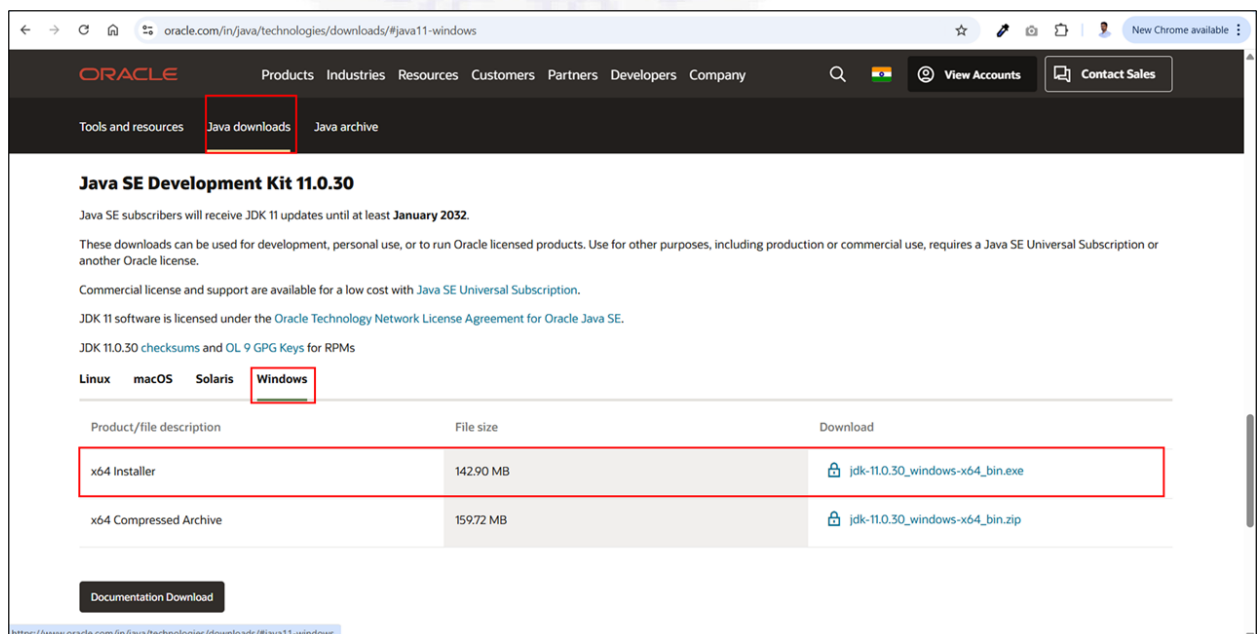
-----
Gradle 9.3.0
-----

Build time:   2026-01-16 11:14:22 UTC
Revision:     701205ed2f78811508466c8e1952304c2ea869f5

Kotlin:       2.2.21
Groovy:       4.0.29
Ant:          Apache Ant(TM) version 1.10.15 compiled on August 25 2024
Launcher JVM: 11.0.29 (Microsoft 11.0.29+7-LTS)
Daemon JVM:   C:\Program Files\Microsoft\jdk-11.0.29-hotspot (no Daemon JVM specified, using current Java home)
OS:           Windows 11 10.0 amd64
```

Install JDK

Download Java JDK from [Java Downloads](#) and install it.



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Java SE Development Kit 11.0.30

Java SE subscribers will receive JDK 11 updates until at least **January 2032**.

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JDK 11.0.30 [checksums](#) and [OL 9 GPG Keys](#) for RPMs

Linux macOS Solaris **Windows**

Product/file description	File size	Download
x64 Installer	142.90 MB	jdk-11.0.30_windows-x64_bin.exe
x64 Compressed Archive	159.72 MB	jdk-11.0.30_windows-x64_bin.zip

[Documentation Download](#)

<https://www.oracle.com/in/java/technologies/downloads/#java11-windows>

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To check the installed Java version, open the Command Prompt and run **java -version**.

```
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thejan>java -version
openjdk version "11.0.29" 2025-10-21 LTS
OpenJDK Runtime Environment Microsoft-12580167 (build 11.0.29+7-LTS)
OpenJDK 64-Bit Server VM Microsoft-12580167 (build 11.0.29+7-LTS, mixed mode, sharing)
```

